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PI Name:	Hu, Jian Zhi Ph.D.		
Project Title:	Early Detection of Inflammatory Response and the Subsequent Health Outcomes Due to High LET Particle Radiation: An Integrated Metabolomics Study		
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Space Biology Special Category:	None		
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Task Description:	Our new studies reveal that NMR-detectable metabolites (e.g., the concentrations of alanine, acetate, glutamate, creatine, choline, phosphocholine, inositol and myo-inositol, etc.) are increased significantly after rats were exposed to whole body high linear energy transfer (LET) iron ions at doses as low as 0.16 Gy. Dramatic changes were also detected in metabolites in various excised organs from mice exposed to gamma-rays at doses of 3.0 and 7.8 Gy. We hypothesize that metabolite biomarkers sensitive to both high LET particle radiation and low LET gamma-rays will be detectable and quantifiable using both established, and unique, high resolution and high sensitivity NMR and LC-MS metabolomics. We also propose that low doses (<=0.5Gy) of high LET radiation causes a low grade inflammatory response and triggers a cascade of molecular events detectable with our novel approach. We will perform an integrated NMR and LC-MS metabolomics study of the impact of exposing mice and rats to 0, 0.5, and 3 Gy using both high LET proton and iron ion radiation. The metabolic perturbations in global physiology and the subsequent health outcomes will be investigated from 1 day to 1 year post irradiation using key body fluids and excised tissues. In parallel, we will investigate the impact of exposing mice and rats to gamma-rays at low and high doses to determine the differences in dose-dependent effects between proton, iron ions and gamma-rays. Furthermore, histopathologic analysis and basic-translational cell and molecular investigations will be used to link metabolic "signatures" and disease pathogenesis. The ultimate goal is to elucidate the detailed molecular mechanisms of high LET radiation, with emphasis on understanding the molecular basis of radiation through the discovery of metabolic signatures. This new knowledge will be translated to benefit human health in space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Histopathologic analysis and basic-translational cell and molecular investigations will be used to link metabolic "signatures" and disease pathogenesis. The ultimate goal is to elucidate the detailed molecular mechanisms of high LET radiation, with emphasis on understanding the molecular basis of radiation through the discovery of metabolic signatures. This new knowledge will be translated to benefit human health in space exploration.

Proposal Title: Early detection of inflammatory responses and subsequent health outcomes due to high LET particle radiation: An integrated metabolomics study
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A. Introduction and Hypotheses

Currently, little is known about the damage to the metabolic pathways of a living system due to radiation in outer space. We hypothesize that (i) the metabolic profile of a living system will be altered as a result of radiation in space; and (ii) the metabolic signatures can be used as early indicators of inflammatory responses that predispose to cancer.

To test these hypotheses, metabolic profiling of tissues and blood were carried out in our one year NASA funded project (10/2007-09/2008) using an integrated, comprehensive NMR and LC-MS metabolomic study of samples obtained from mice, separately exposed to high LET iron ion (56Fe), proton (1H), and low LET gamma-radiation, respectively. In parallel, histopathologic analysis and basic-translational cell and molecular investigations were used to link metabolic "signatures" inflammation and tissue damage. In this report, the major findings resulted from our project are reported.

B. Approaches

Whole-body space radiation experiments: Adult female C57BL/6 and BALB/c mice of age ~ 8 weeks old were exposed to a single dose of either 0 or 3.0 Gy high LET 56Fe (1000MeV/nucleon, 1 Gy/min), or 3.0 Gy 1H (1000 MeV/nucleon, 0.2 Gy/min), or a combined 3.0 Gy 56Fe and 3.0 Gy 1H radiation using the Alternating Gradient Synchrotron (AGS) at NSRL/NASA of Brookhaven National Laboratory. Tissues (femur, spleen, and GI tract) and blood (serum) samples were harvested 4 and 11 days post exposure and subsequently analyzed by NMR and LC-MS for metabolomics, as well as histopathology and basic-translational studies.

Whole-body gamma-irradiation and sample collection. Female C57BL/6 and BALB/c mice of 7 weeks of age were purchased from the Jackson Laboratory. After acclimation for one week at the animal facility of Pacific Northwest National Laboratory (PNNL), they were randomly grouped before being exposed to whole body gamma-radiation using a high activity gamma sources (1250 keV 60Co). All animal work was approved by the Institutional Animal Care and Use Committee (IACUC) at both BNL and PNNL.

C. Major Results

C1: Application of High-Resolution 1H MAS NMR Spectroscopy to the Analysis of Intact Bones from Mice Exposed to Whole-body Gamma-Radiation

We have demonstrated that high resolution magic angle spinning (hr-MAS) 1H NMR can be successfully utilized to rapidly profile the pathology of bone marrow with minimal sample preparation. The spectral resolution obtained allows many metabolites to be analyzed quantitatively. It is found that the level of NMR detectable metabolites in the epiphysis + metaphysis sections of mouse femur is statistically significantly higher than that observed in the diaphysis of the same femur. The major metabolite damages to bone marrow resulting from either 3.0 Gy or 7.8 Gy of whole-body gamma-radiation, especially at early times, i.e., 4 days post exposure, were (i) decreased total choline content, (ii) increased fatty acids in bone marrow, and (iii) decreased creatine content. These results suggest that the membrane choline phospholipid metabolism (MCPM) pathway, the basic cellular energy pathway are altered as a result of radiation. It is also found that the metabolic damage of radiation to the epiphysis + metaphysis sections of mouse femur is higher than that of the diaphysis of the same femur. Traditional histopathology analysis was also carried out to correlate the impact of radiation damage with metabolite changes. The molecular information from hr-MAS 1H NMR complements the pathology data.

C2: Application of High-Resolution 1H MAS NMR Spectroscopy to the Analysis of Intact Bones from Mice Exposed to Whole-body high LET Iron Ion and Proton Radiation

We found that the relative amount of mobile lipids and glycerol esters in the femurs of mice exposed to 3.0 Gy 56Fe or 3.0 Gy 1H 4 days post-exposure were significantly ($p<0.05$) increased compared with controls. At 11 days post-exposure, the amount of mobile lipids was further increased, and the increase arising from 1H radiation was more than that from 56Fe radiation. This result indicates that using the same radiation dose, 1H radiation had more sustained

Task Progress:	effects than 56Fe radiation. It was also found that the amount of total choline (choline + phosphocholine + glycerophosphocholine) and that the glycogen/glucose tended to recover to the level of control animals at 11 days post exposure in the femurs of mice receiving either the 3.0 Gy 56Fe or 3.0 Gy 1H. For the mice receiving the combined 3.0 Gy 56Fe and 3.0 Gy 1H exposure, the increase of glycerol ester was additive of the damages from the individual exposures, but synergetic effects were observed in lipids. Furthermore, the relative amount of glycogen/glucose was significantly increased while the relative amount of total choline was significantly decreased in the femurs of mice that received the combined 56Fe and 1H exposure. These results are similar to those obtained from mice exposed to whole-body gamma radiation and at similar radiation doses. However, it was noticed that the metabolic damage is higher in the case of space radiation than that of gamma radiation at similar radiation doses. C3: Application of LC-MS metabolomics to the Analysis of Bone Marrow and Blood from Mice Exposed to Whole-body Iron ion and proton Radiation LC-MS metabolomics study further reveals that more than 120 lipid features were changed significantly ($P < 0.01$) at 4 days post radiation. In particular, the control, 3.0 Gy 56Fe, 3.0 Gy 1H, and the combined 3.0 Gy 56Fe and 3.0 Gy 1H treatment (named as 6.0 Gy combined dose) groups can be clearly distinguished from each other by principle component analysis. Among these 120 lipid features, the chemical compositions of 40 lipids were identified based on our home-built lipid Accurate Mass and Time tag library, including membrane choline lipids such as Glycerophosphocholine, Lysoglycerophosphocholine, etc., and other lipids such as Sphingomyelin, Triacylglycerol, etc. We have found that compared with the control group, the level of PC in bone marrow decreased statistically and significantly by a factor of about 24 folds in the combined 3.0 Gy 56Fe and 3.0 Gy proton radiation group. This decreasing factor is about 8 in the 3.0 Gy 56Fe, and about 2.7 in the 3.0 Gy proton radiated group. These results confirm the results obtained from NMR-metabolomics studies above. Therefore, it can be concluded that PC can be utilized as a molecular biomarker characterizing radiation dose and type. The PC may also be a potential molecular marker than can be utilized at radiation doses as low as less than 0.5 Gy given the large decrease at currently used high radiation doses. C4: Major histopathology findings from the bone marrow samples are as follows. 56Fe 3Gy, 4 days: depletion, diffuse, subacute, moderate; 56Fe 3Gy, 11 days: normal to minimally hyperplastic, diffuse, subacute; 1H 3Gy, 4 days: normal to slight decrease in cellularity, diffuse, subacute; 1H 3Gy, 11 days: normal; Combined 3.0 Gy 56Fe and 3.0 Gy 1H, 4 days: depletion, diffuse, subacute, severe. Similar results were found from the bone marrow of mice exposed to whole-body gamma-radiation. C5: Basic-translational studies. Basic-translational studies using Luminex (XMAP), ELISA, western blotting on serum samples indicate that an important mediator of inflammation and thrombosis called CD40L, a member of the tumor necrosis factor family, showed the largest change. Compared with the results from control animals, the level of sCD40L increased at 4 days post 3.0 Gy 56Fe radiation, followed by a decrease at 11 days post radiation. In contrast, the level of sCD40L decreased at 4 days post 3.0 Gy 1H radiation, followed by an increase at 11 days post radiation. For the combined 3.0 Gy 56Fe and 3.0 Gy 1H radiation 4 days post exposure, the level of sCD40L is approximately the average of the individual exposure. D. Conclusion Metabolomics provide molecular level information that can not be obtained otherwise from traditional histopathological and basic-translational investigations. Metabolite biomarkers characterizing radiation damage can be identified with a combination of advanced NMR and LC-MS metabolomics. Based on NMR metabolomics, we have found that a group of metabolite related to the membrane choline phospholipid metabolism (MCPM) pathway, the basic cellular energy pathway are altered as a result of radiation. While NMR metabolomics is more quantitative than LC-MS metabolomics, LC-MS metabolomics has much higher sensitivity and resolution, allowing a large range of metabolites to be identified. In particular, LC-MS metabolomics is capable of differentiating lipids with subtle mass differences, thus allowing both up-, and down-regulation of various lipids identified simultaneously. We have found that hundreds of lipid features are statistically and significantly changed in the bone marrow of mice exposed to high LET space radiation at currently investigated radiation dose of 3.0 Gy 56Fe, 3.0 Gy proton, or the combined 3.0 Gy 56Fe and 3.0 Gy proton whole-body radiation. Among these features, we have already found that phosphocholine, a known cancer biomarker, is a potential metabolite biomarker that can be utilized for risk assessment in space radiation. We are currently preparing three research papers based on the results obtained from our one year grant and the details are given below. 1. Zhang Q, Hu JZ, Rommereim DN, Murphy MK, Phipps RP, Huso DL, and Dicello JF. Application of high-resolution 1H MAS NMR spectroscopy to the analysis of intact bones from mice exposed to gamma-Radiation. Magnetic Resonance in Medicine. To be submitted early 2009. 2. Zhang Q, Hu JZ, Rommereim DN, Isern NG, Metz TO, Murphy MK, Huso DL, Dicello JF, and Phipps RP. Probing the metabolic changes in mice exposed to high LET iron ion and proton radiation via NMR metabolomics. Manuscript in preparation. 3. Zhang Q, Hu JZ, Rommereim DN, Metz TO, Murphy MK, Huso DL, Dicello JF, and Phipps RP. Investigation of the metabolic changes in bone marrow from mice exposed to high LET iron ion and proton radiation using a combination of LC-MS metabolomic and histopathology approach. Manuscript in preparation.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
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Abstracts for Journals and Proceedings	Hu JZ, Zhang Q, Rommereim DN, Isern NG, Metz TO, Murphy MK, Huso DL, Dicello JF, Phipps RP. "Early detection of inflammatory responses and subsequent health outcomes due to high LET particle radiation: an integrated metabolomics study." To be presented at NASA Human Research Program Investigators' Workshop, South Shore Harbour Resort and Conference Center, League City, Texas, February 2-4, 2009. NASA Human Research Program Investigators' Workshop, South Shore Harbour Resort and Conference Center, League City, Texas, February 2-4, 2009. , Feb-2009
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